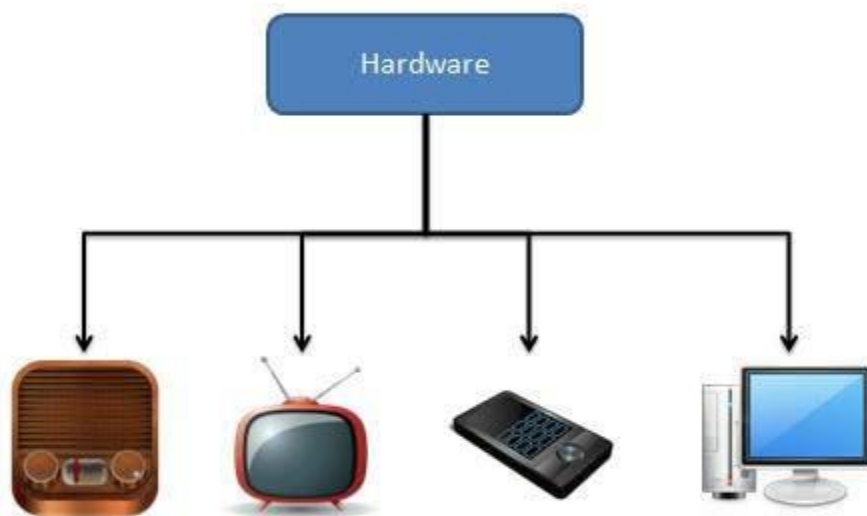


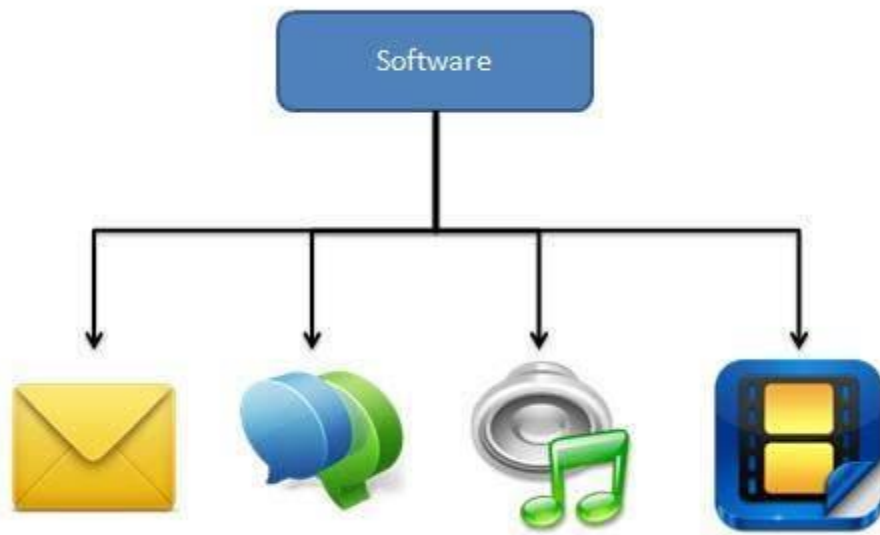
# UNIT 7 Contemporary Technologies

## Definition of Multimedia

By definition Multimedia is a representation of information in an attractive and interactive manner with the use of a combination of text, audio, video, graphics and animation. In other words we can say that Multimedia is a computerized method of presenting information combining textual data, audio, visuals (video), graphics and animations. For examples: E-Mail, Yahoo Messenger, Video Conferencing, and Multimedia Message Service (MMS).

Multimedia as name suggests is the combination of Multi and Media that is many types of media (hardware/software) used for communication of information.





## Components of Multimedia

Following are the common components of multimedia:

- **Text-** All multimedia productions contain some amount of text. The text can have various types of fonts and sizes to suit the professional presentation of the multimedia software.
- **Graphics-** Graphics make the multimedia application attractive. In many cases people do not like reading large amount of textual matter on the screen. Therefore, graphics are used more often than text to explain a concept, present background information etc. There are two types of Graphics:
  - **Bitmap images-** Bitmap images are real images that can be captured from devices such as digital cameras or scanners. Generally bitmap images are not editable. Bitmap images require a large amount of memory.
  - **Vector Graphics-** Vector graphics are drawn on the computer and only require a small amount of memory. These graphics are editable.
- **Audio-** A multimedia application may require the use of speech, music and sound effects. These are called audio or sound element of multimedia. Speech is also a perfect way for teaching. Audio are of analog and digital types. Analog audio or sound refers to the original sound signal. Computer stores the sound in digital form. Therefore, the sound used in multimedia application is digital audio.
- **Video-** The term video refers to the moving picture, accompanied by sound such as a picture in television. Video element of multimedia application gives a lot of information in small duration of time. Digital video is useful in multimedia application for showing real life objects. Video have highest performance demand on the computer memory and on the bandwidth if placed on the internet. Digital video files can be stored like any other files in the computer and the quality of the

video can still be maintained. The digital video files can be transferred within a computer network. The digital video clips can be edited easily.

- **Animation-** Animation is a process of making a static image look like it is moving. An animation is just a continuous series of still images that are displayed in a sequence. The animation can be used effectively for attracting attention. Animation also makes a presentation light and attractive. Animation is very popular in multimedia application

## Applications of Multimedia

Following are the common areas of applications of multimedia.

- **Multimedia in Business-** Multimedia can be used in many applications in a business. The multimedia technology along with communication technology has opened the door for information of global work groups. Today the team members may be working anywhere and can work for various companies. Thus the work place will become global. The multimedia network should support the following facilities:
  - Voice Mail
  - Electronic Mail
  - Multimedia based FAX
  - Office Needs
  - Employee Training
  - Sales and Other types of Group Presentation
  - Records Management
- **Multimedia in Marketing and Advertising-** By using multimedia marketing of new products can be greatly enhanced. Multimedia boost communication on an affordable cost opened the way for the marketing and advertising personnel. Presentation that have flying banners, video transitions, animations, and sound effects are some of the elements used in composing a multimedia based advertisement to appeal to the consumer in a way never used before and promote the sale of the products.
- **Multimedia in Entertainment-** By using multimedia marketing of new products can be greatly enhanced. Multimedia boost communication on an affordable cost opened the way for the marketing and advertising personnel. Presentation that have flying banners, video transitions, animations, and sound effects are some of the elements used in composing a multimedia based advertisement to appeal to the consumer in a way never used before and promote the sale of the products.
- **Multimedia in Education-** Many computer games with focus on education are now available. Consider an example of an educational game which plays various rhymes for kids. The child can paint the pictures, increase reduce size of various objects etc apart from just playing the rhymes. Several other multimedia packages are available in the market which provide a lot of detailed information and playing capabilities to kids.
- **Multimedia in Bank-** Bank is another public place where multimedia is finding more and more application in recent times. People go to bank to open saving/current

accounts, deposit funds, withdraw money, know various financial schemes of the bank, obtain loans etc. Every bank has a lot of information which it wants to impart to its customers. For this purpose, it can use multimedia in many ways. Bank also displays information about its various schemes on a PC monitor placed in the rest area for customers. Today on-line and internet banking have become very popular. These use multimedia extensively. Multimedia is thus helping banks give service to their customers and also in educating them about banks attractive finance schemes.

- **Multimedia in Hospital-** Multimedia best use in hospitals is for real time monitoring of conditions of patients in critical illness or accident. The conditions are displayed continuously on a computer screen and can alert the doctor/nurse on duty if any changes are observed on the screen. Multimedia makes it possible to consult a surgeon or an expert who can watch an ongoing surgery line on his PC monitor and give online advice at any crucial juncture.

In hospitals multimedia can also be used to diagnose an illness with CD-ROMs/ Cassettes/ DVDs full of multimedia based information about various diseases and their treatment. Some hospitals extensively use multimedia presentations in training their junior staff of doctors and nurses. Multimedia displays are now extensively used during critical surgeries.

- **Multimedia Pedagogues-** Pedagogues are useful teaching aids only if they stimulate and motivate the students. The audio-visual support to a pedagogue can actually help in doing so. A multimedia tutor can provide multiple numbers of challenges to the student to stimulate his interest in a topic. The instruction provided by pedagogue have moved beyond providing only button level control to intelligent simulations, dynamic creation of links, composition and collaboration and system testing of the user interactions.

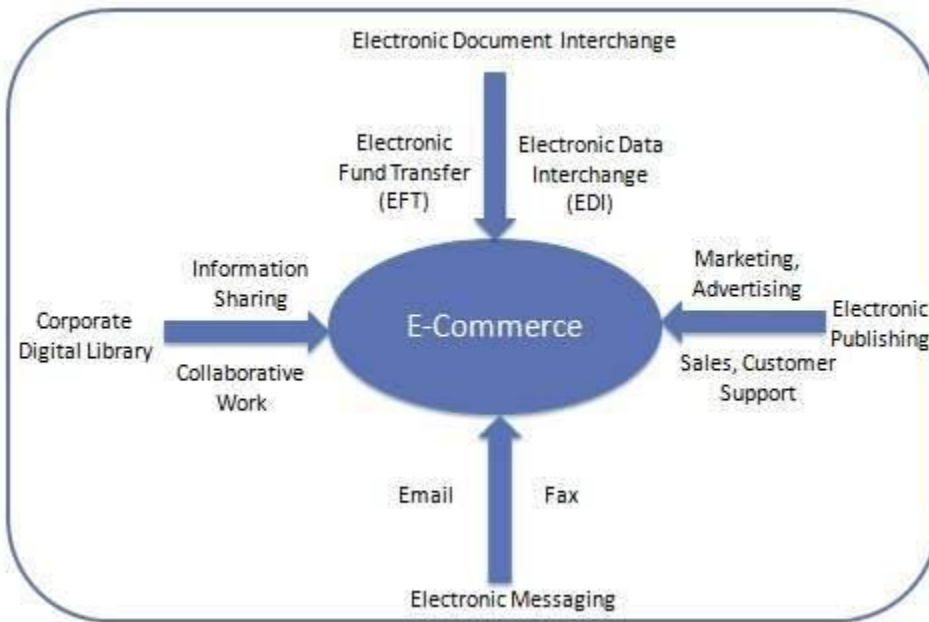
- **Communication Technology and Multimedia Services-** The advancement of high computing abilities, communication ways and relevant standards has started the beginning of an era where you will be provided with multimedia facilities at home. These services may include:

- Basic Television Services
- Interactive entertainment
- Digital Audio
- Video on demand
- Home shopping
- Financial Transactions
- Interactive multiplayer or single player games
- Digital multimedia libraries
- E-Newspapers, e-magazines

**E-Commerce** or Electronics Commerce is a methodology of modern business, which addresses the need of business organizations, vendors and customers to reduce cost and improve the quality of goods and services while increasing the speed of delivery.

Ecommerce refers to the paperless exchange of business information using the following ways –

- Electronic Data Interchange (EDI)
- Electronic Mail (e-mail)
- Electronic Bulletin Boards
- Electronic Fund Transfer (EFT)
- Other Network-based technologies



## Features

E-Commerce provides the following features –

- **Non-Cash Payment** – E-Commerce enables the use of credit cards, debit cards, smart cards, electronic fund transfer via bank's website, and other modes of electronics payment.
- **24x7 Service availability** – E-commerce automates the business of enterprises and the way they provide services to their customers. It is available anytime, anywhere.
- **Advertising / Marketing** – E-commerce increases the reach of advertising of products and services of businesses. It helps in better marketing management of products/services.
- **Improved Sales** – Using e-commerce, orders for the products can be generated anytime, anywhere without any human intervention. It gives a big boost to existing sales volumes.

- **Support** – E-commerce provides various ways to provide pre-sales and post-sales assistance to provide better services to customers.
- **Inventory Management** – E-commerce automates inventory management. Reports get generated instantly when required. Product inventory management becomes very efficient and easy to maintain.
- **Communication improvement** – E-commerce provides ways for faster, efficient, reliable communication with customers and partners.

## Traditional Commerce v/s E-Commerce

| Sr. No. | Traditional Commerce                                                                                                           | E-Commerce                                                                                                                                                                   |
|---------|--------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | Heavy dependency on information exchange from person to person.                                                                | Information sharing is made easy via electronic communication channels making little dependency on person to person information exchange.                                    |
| 2       | Communication/ transaction are done in synchronous way. Manual intervention is required for each communication or transaction. | Communication or transaction can be done in asynchronous way. Electronics system automatically handles when to pass communication to required person or do the transactions. |
| 3       | It is difficult to establish and maintain standard practices in traditional commerce.                                          | A uniform strategy can be easily established and maintain in e-commerce.                                                                                                     |
| 4       | Communications of business depends upon individual skills.                                                                     | In e-Commerce or Electronic Market, there is no human intervention.                                                                                                          |
| 5       | Unavailability of a uniform platform as traditional commerce depends heavily on personal communication.                        | E-Commerce website provides user a platform where all information is available at one place.                                                                                 |
| 6       | No uniform platform for information sharing as it depends heavily on personal communication.                                   | E-Commerce provides a universal platform to support commercial / business activities across the globe.                                                                       |

## **Advantages of e-commerce**

### **Advantages to Organizations**

- Using e-commerce, organizations can expand their market to national and international markets with minimum capital investment. An organization can easily locate more customers, best suppliers, and suitable business partners across the globe.
- E-commerce helps organizations to reduce the cost to create process, distribute, retrieve and manage the paper based information by digitizing the information.
- E-commerce improves the brand image of the company.
- E-commerce helps organization to provide better customer services.
- E-commerce helps to simplify the business processes and makes them faster and efficient.
- E-commerce reduces the paper work.
- E-commerce increases the productivity of organizations. It supports "pull" type supply management. In "pull" type supply management, a business process starts when a request comes from a customer and it uses just-in-time manufacturing way.

### **Advantages to Customers**

- It provides 24x7 support. Customers can enquire about a product or service and place orders anytime, anywhere from any location.
- E-commerce application provides users with more options and quicker delivery of products.
- E-commerce application provides users with more options to compare and select the cheaper and better options.
- A customer can put review comments about a product and can see what others are buying, or see the review comments of other customers before making a final purchase.
- E-commerce provides options of virtual auctions.
- It provides readily available information. A customer can see the relevant detailed information within seconds, rather than waiting for days or weeks.
- E-Commerce increases the competition among organizations and as a result, organizations provides substantial discounts to customers.

## Advantages to Society

- Customers need not travel to shop a product, thus less traffic on road and low air pollution.
- E-commerce helps in reducing the cost of products, so less affluent people can also afford the products.
- E-commerce has enabled rural areas to access services and products, which are otherwise not available to them.
- E-commerce helps the government to deliver public services such as healthcare, education, social services at a reduced cost and in an improved manner.

## Disadvantages of e-commerce

### Technical Disadvantages

- There can be lack of system security, reliability or standards owing to poor implementation of e-commerce.
- The software development industry is still evolving and keeps changing rapidly.
- In many countries, network bandwidth might cause an issue.
- Special types of web servers or other software might be required by the vendor, setting the e-commerce environment apart from network servers.
- Sometimes, it becomes difficult to integrate an e-commerce software or website with existing applications or databases.
- There could be software/hardware compatibility issues, as some e-commerce software may be incompatible with some operating system or any other component.

### Non-Technical Disadvantages

- **Initial cost** – The cost of creating/building an e-commerce application in-house may be very high. There could be delays in launching an e-Commerce application due to mistakes, and lack of experience.
- **User resistance** – Users may not trust the site being an unknown faceless seller. Such mistrust makes it difficult to convince traditional users to switch from physical stores to online/virtual stores.
- **Security/ Privacy** – It is difficult to ensure the security or privacy on online transactions.



- Lack of touch or feel of products during online shopping is a drawback.
- E-commerce applications are still evolving and changing rapidly.
- Internet access is still not cheaper and is inconvenient to use for many potential customers, for example, those living in remote villages.

## **E-learning**

E-Learning is the process of sharing knowledge through various channels such as e-books, CDs, webinars, and more.

It has revolutionized the conventional method of chalk and board style of learning imparted to the students.

Unlike this, e-Learning education makes giving and receiving simpler, more prolific, and productive. Tutors apply the method of teaching purely through the latest technology.

We define E-Learning as Courses that are specifically delivered through the internet everywhere other than the traditional classroom where the teacher is teaching. E-learning refers to the use of the internet, technology, and acquired knowledge and Skills.

Thus, teaching and learning both become simpler, easier, and more effective.

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## Importance of e-learning

- E-learning has been introduced to empower learners to get basic schooling and enhance skills. Also, they can obtain a degree certificate, without actually attending school or university or any other institution.
- For tutors, it is a great source of earning, wherein they can teach from anywhere in their preferred time.
- Applying e-learning to all levels of schooling helped to ensure students grasp the lessons adequately at a faster pace.
- According to psychology, the audio-visual method of teaching leads to a disciplined learning environment. There are effective tutor and student engagements.
- One of the **importance of e-learning in education** is that tutors and participants both can develop advanced learning skills. For example, creating and selling eBooks is one such advancement.
- E-learning has worked towards bringing learners, tutors, experts, practitioners, and other interest groups to one place. Thus, there is a good practice of knowledge sharing followed through different online platforms. This is important in current times as competition is rising and the world is also growing. Hence, quick information helps in the better growth of an individual.

Digital and self-initiated learning can be acquired at the desired locations. One need not wander in search of learning as e-learning

education is versatile and amply equipped to suit all learning methods.

E-Learning is conducted and eventuates in many forms and at times is the blend of the following methodologies and practices:

- Completed online – no face-to-face meetings
- Provided through Mixed Learning – A blend of online and direct communication

### **That is in the two different forms**

1. Synchronous- the communication between the provider and the receiver happens directly as in chat rooms, or video-audio conferencing.
2. Asynchronous- The information is passed through forums, emails, wikis, etc.
3. It is promoted through self-study practices also.
4. Web-based learning is another better option
5. CD-ROMs provide subject-wise detailed learning
6. Audio and Visuals are supplementary, descriptive, and communicative methods.

## **Benefits of E-Learning in Education**

An analytical look at the benefits of online learning will make us consider how today's teachers, students, and institutes want to get relevant and easily accessible content.

This is fulfilled with the online mode of learning as the students can learn at their comfort and requirement.

Also, teachers can teach with much more flexibility.

- Online Learning can accommodate everyone's needs

- Classes can be taken from any place and at the time that students or tutors prefer.
- It offers access to exclusive, prolific, and updated content and accessibility is open, secure, and uninterrupted.
- E-Learning lets you be in sync with modern learners and updated with the current trends
- It ensures quick delivery of lessons. Traditional classrooms involve some kind of delay. Whereas, whereas eLearning provides expeditious and exclusive delivery of lessons. There is no procrastinator in eLearning. It is a quick way of learning!
- The scalability of learning, content, and duration that is taken can be reasonably measured. It is beneficial to those who feel nervous and disconnected in groups. It helps one learn without having to give up the comforts of the environment. You are at ease with Consistency of exposure, inputs, results, and coordination is highly ranked in E-Learning as it allows teachers a higher degree of coverage to deliver the content regularly. This ensures consistency in learning.
- The method is made available at much-reduced costs. The importance of E-Learning education is that it is quick and does not require much cost. The long training period, infrastructure, stationery, travel expenses, etc. is reduced.
- The effectiveness of the transferred or imparted knowledge and learning is high and powerful. It makes information easy to grasp and absorb. The Audio-Visuals help in remembering knowledge for a longer time. Also, the courses which tutors prepare are well-planned.
- Mobility assured is a very comfortable and affordable option. Taking revision and getting hands-on with different courses is not that simple in traditional classes. Contrary to that, missed lessons can always be taken again online. This makes it easier for tutors to impart the right information.

- It promotes a self-paced learning process.

Through e-learning, students can develop a self-paced training schedule. Also, for tutors, things become quite easy. People working 9 to 5 or those who simply don't have the necessary time to conduct physical classes are no longer left behind.

They can take online sessions after work, in the evening, on the weekends, or whenever they are free.

All they need is a proper internet facility, computer or laptop, microphones, and headphones. E-Learning has a vast amount of use and also helps students in education.

- Tutors have a variety of options to teach such as you can create recorded lessons, conducting short webinars, and selling courses with yearly packages. Hence, there are numerous possibilities for earning a side income or even a regular income.

## **Benefits of E-Learning For Students**

Unlike the traditional method of teaching with board and chalk, E-learning is fulfilled with the online mode of learning as the students can learn at their comfort and requirements.

Let's have a look at the advantages of E-Learning for the students:

- You can access the study material unlimited times.
- You can study your Courses anytime and anywhere.
- Students can access updated content when they want.
- Unlike the traditional method of teaching, e-learning has a quick mode of delivery. This indicates that learning time is reduced.
- E-Learning provides scalability which helps in providing training.

- All students can receive the same type of syllabus, study materials, and train through E-Learning.
- Through E-Learning, you can save time, and money and reduced transportation costs. so, E-Learning is cost-effective compared to traditional learning.
- E-Learning is provided online so No need for papers like traditional learning. Thus, we can save the Environment through this.

One can link the various resources in several varying formats.

- Web-based learning promotes active and independent learning.
- College students, people working full time, homeowners, YouTubers, and coaching institutes can take advantage of **elearning education**.
- One biggest advantage of the internet or e-learning is that one can educate oneself in the comfort of their own home and get a degree.

These are some impossible things that are made possible only because of technology.

- Another benefit is that eLearning is based on convenience and flexibility. All the resources for the students as well as teachers are available in one place.
- Anyone can get training on a day-to-day basis. It can be on weekends or whenever one has free time.
- E-learning is also beneficial for business. This is because employees should also be educated about new skills. It helps in improving their efficiency and is fruitful for the future [education course](#) of time.

- There is an easy way for clearing doubts through discussion boards or chat boxes. Tutors can easily answer the queries of students. Thus, it leads to better interaction.
- Global level education- Tutors can provide online education in multiple languages and to people from different time zones.
- Last but not least eLearning is a boon to the environment as paperless teaching has contributed a lot to saving trees. Also, tutors are feeling this option is great as now the evaluation of each student is done online which is quicker and less tiring.

## **E-governance**

Electronic Governance is the application of Information and Communication Technologies (ICTs) for delivering government services through integration of various stand-alone systems between Government-to-Citizens (G2C), Government-to-Business (G2B), and Government-to-Government(G2G) services. It is often linked with back-office processes and interactions within the entire government framework. Through **e-Governance**, the government services are made available to the citizens in a convenient, efficient, and transparent manner.

## **Advantages of E-Governance**

### **Speed**

[Technology](#) makes communication swifter. Internet, smartphones have enables instant transmission of high volumes of data all over the world.

### **Saving Costs**

A lot the Government expenditure goes towards the cost of buying stationery for official purposes. Letters and written records consume a lot of stationery. However, replacing them with [smartphones](#) and the internet can saves crores of money in expenses every year.

### **Transparency**

The use of e-governance helps make all functions of the business transparent. All Governmental information can be uploaded onto the internet. The citizens access specifically access whichever information they want, whenever they want it, at the click of a mouse, or the touch of a finger.

However, for this to work the Government has to ensure that all data as to be made public and uploaded to the Government information forums on the internet.

## **Accountability**

Transparency directly links to [accountability](#). Once the functions of the government are available, we can hold them accountable for their actions.

## **Disadvantages of E-Governance**

### **Loss of Interpersonal Communication**

The main disadvantage of e-governance is the loss of interpersonal communication. Interpersonal communication is an aspect of communication that many people consider vital.

### **High Setup Cost and Technical Difficulties**

Technology has its disadvantages as well. Specifically, the setup cost is very high and the machines have to be regularly maintained. Often, computers and [internet](#) can also break down and put a dent in governmental work and services.

### **Illiteracy**



A large number of people in India are illiterate and do not know how to operate computers and smartphones. E-governance is very difficult for them to access and understand.

## **Cybercrime/Leakage of Personal Information**

There is always the risk of private data of citizens stored in government servers being stolen. Cybercrime is a serious issue, a breach of data can make the public lose confidence in the Government's ability to govern the people.

E-Governance is of 4 types depending on the specific types of services.

### **1. Government-to-Citizen(G2C)**

The Government-to-citizen refers to the government services that are accessed by the familiar people. And Most of the government services fall under G2C. Likewise, the primary goal of Government-to-citizen is to provide facilities to the citizen. It helps the ordinary people to reduce the time and cost to conduct a transaction. A citizen can have access to the services anytime from anywhere.

Furthermore, Many services like license renewals, and paying tax are essential in G2C. Likewise, spending the administrative fee online is also possible due to G2C. The facility of Government-to-Citizen enables the ordinary citizen to overcome time limitation. It also focuses on geographic land barriers.

## **2. Government-to-business (G2B)**

The Government to business is the exchange of services between Government and Business organizations. It is efficient for both government and business organizations. G2B provides access to relevant forms needed to comply. The G2B also consists of many services exchanged between business sectors and government.

Similarly, the Government to business provides Timely business information. And A business organization can have easy and convenient online access to government agencies. G2B plays a crucial role in business development. It enhances the efficiency and quality of communication and transparency of government projects.

## **3. Government-to-Government (G2G)**

The Government-to-Government refers to the interaction between different government department, organizations, and agencies. This increases the efficiency of government processes. In G2G, government agencies can share the same database using online communication. The government departments can work together. This service can increase international diplomacy and relations.

In conclusion, G2G services can be at the local level or the international level. It can communicate with global government and local government as well. Likewise, it provides safe and secure inter-relationship between domestic or foreign government. G2G constructs a universal database for all member states to enhance service.

## 4. Government-to-Employee (G2E)

The Government-to-Employee is the internal part of G2G sector. Furthermore, G2E aims to bring employees together and improvise knowledge sharing.

Similarly, G2E provides online facilities to the employees. Likewise, applying for leave, reviewing salary payment record. And checking the balance of holiday. The G2E sector provides human resource training and development. So, G2E is also the relationship between employees, government institutions, and their management.

**E-banking** is a blanket term used to indicate a process through which a customer is allowed to carry out, personal or commercial banking transactions using electronic and telecommunication network.

It is a product offered by [banks](#) which facilitates online banking, with the help of which the customer can have access to the bank account in just one click.

E-banking covers facilities such as – fund transfer, checking account statements, utility bill payments, opening of bank account, locating nearest ATM, obtain information on financial products and services, applying for loans, etc. using a personal computer, smartphone, laptop or personal digital assistant.

### E-banking Services

In simple words, e-banking refers to a banking arrangement, with which the customer can perform various transactions over the internet, which is end-to-end encrypted, i.e. it is completely safe and secure.

E-banking promotes paperless/cashless transactions. It comes with a number of rights, responsibilities and fees as well. The range of services covered under E-banking are:

1. Internet Banking: A banking facility provided to the customers through which the customers are able to perform a number of monetary and non-monetary transactions, using the internet, through the bank's website or application.
2. Mobile Banking: Almost all the banks have designed their mobile applications with which you can perform transactions at your fingertips. For this, four things are required –

a smartphone, internet, mobile application, and mobile banking service enabled in your bank account.

3. **ATM: Automated Teller Machine**, popularly known as ATM is one of the most common and initial service, provided under e-banking. It is not just a machine with which you can withdraw cash as and when required, but it also allows you to check your account status, transfer fund, deposit fund, changes mobile number, change Debit Card PIN, i.e. Personal Identification Number.
4. **Debit Card**: Debit cards are used in our day to day life so as to perform end number of transactions. Debit cards are linked to the customer's bank account and so the customer only needs to swipe the card, in order to make payment at Point of Sale (POS) outlets, online shopping, ATM withdrawal. In this way, the amount is deducted from the customer's account directly.
5. **Credit Card**: Just like a debit card, a credit card is also a payment card which the banks issue to the customers on their request, after checking their credit score and history. It enables the cardholder to borrow funds upto the pre-approved limit and make payment. The limit is granted by the banks which issue the card. The cardholder promises to repay the amount within a stipulated time, with some charges, for the use of credit card.
6. **Point of Sale (POS)**: Points of sale system refers to the point, in terms of date, time and place (retail outlet) where the customer makes a payment, using a plastic card, for the purchase made or services received.
7. **Electronic Data Interchange (EDI)**: EDI is a new mode of communicating information between businesses electronically using a standardized format, which was conventionally paper-based.
8. **Electronic Fund Transfer (EFT)**: When money is transferred electronically from one bank to another, it is called as electronic fund transfer. It covers direct debit, direct deposits, wire transfers etc.

### **Benefits of E-banking**

- It enables digital payments, which encourages transparency.
- It allows 24/7 access to the bank account.
- It also sends notifications and alerts to get updated with the banking transactions and changes in the rules.
- It lowers transaction cost for the banks.
- It is convenient and easy for customers, as they are not required to visit the bank branch every time.

In a nutshell, any type of banking transaction performed through electronic mode comes under E-banking.

It is a secure, fast and convenient electronic banking facility that allows its customers to undertake online banking services anytime during the day and at any place using the internet, for which the customers used to visit the banks in earlier days.

# Hypermedia

- It refers to connecting the hypertext with media such as graphics, sounds, and animations.
- It involves graphics, image, video, and audio.
- It can be understood as the improved version of hypertext.
- Text with multimedia is a part of the link.
- It allows the user to click on the text or any other multimedia to move from one page to another page.
- It gives flexibility of movement.
- It attracts more number of users.
- It provides a better user experience.
- Example includes reading an article on a website, and click on an image takes the user to its associated page.

**A geographic information system (GIS)** is a computer system for capturing, storing, checking, and displaying data related to positions on Earth's surface. By relating seemingly unrelated data, GIS can help individuals and organizations better understand spatial patterns and relationships.

GIS technology is a crucial part of spatial data infrastructure, which the White House defines as “the technology, policies, standards, human resources, and related activities necessary to acquire, process, distribute, use, maintain, and preserve spatial data.”

GIS can use any information that includes location. The location can be expressed in many different ways, such as latitude and longitude, address, or ZIP code.

Many different types of information can be compared and contrasted using GIS. The system can include data about people, such as population, income, or education level. It can include information about the landscape, such as the location of streams, different kinds of vegetation, and different kinds of soil. It can

include information about the sites of factories, farms, and schools, or storm drains, roads, and electric power lines.

With GIS technology, people can compare the locations of different things in order to discover how they relate to each other. For example, using GIS, a single map could include sites that produce pollution, such as factories, and sites that are sensitive to pollution, such as wetlands and rivers. Such a map would help people determine where water supplies are most at risk.

The advantages of using a geographic information system include:

- Improved decision making – decisions are made easier because specific and detailed information is presented about one or more locations.
- Reduce costs and increase efficiency – especially regarding maintenance schedules, fleet movements or scheduling timetables.
- Improved communication between any involved organisations or departments as the visual format is easily understood by all.
- Easy recordkeeping – geographical changes are easily recorded by GIS for those responsible of recording the changes.
- Managing geographically – knowing what is and will be occurring in a geographic space in order to plan a course of action.

## **Application of GIS**

It allows people to see the world in a different way by mapping the position and quantity of things, mapping the density of people and objects and mapping any changes that occur. GIS also allows us to find out what is happening inside a specific area or nearby to a specific area.

More specifically, GIS is used through:

- Environmental Geography – to analyse the impact people have on the environment.

- Physical Geography – to study the elements of atmosphere, biosphere and geosphere.
- Emergency Management Information System – to give real time data to emergency responders about the geographical layout.
- Health Geography – to use geographical information to study health related issues such as disease and illness.
- Economic Geography – to study economic activities across the earth.
- Transportation Geography – to investigate the spatial interactions of people or things.

## What Is Virtual Reality?

Virtual reality is the use of computer technology to create simulated environments. Virtual reality places the user inside a three-dimensional experience. Instead of viewing a screen in front of them, users are immersed in and interact with 3D worlds.

Simulation of human senses—all five of them—transforms a computer into a vehicle into new worlds. The only limitation to a superb VR experience is computing power and content availability.

## **The three main VR categories are the following:**

- Non-Immersive Virtual Reality: This category is often overlooked as VR simply because it's so common. Non-immersive VR technology features a computer-generated virtual environment where the user simultaneously remains aware and controlled by their physical environment. Video games are a prime example of non-immersive VR.
- Semi-Immersive Virtual Reality: This type of VR provides an experience partially based in a virtual environment. This type of VR makes sense for educational and training

purposes with graphical computing and large projector systems, such as flight simulators for pilot trainees.

- **Fully Immersive Virtual Reality:** Right now, there are no completely immersive VR technologies, but advances are so swift that they may be right around the corner. This type of VR generates the most realistic simulation experience, from sight to sound to sometimes even olfactory sensations. Car racing games are an example of immersive virtual reality that gives the user the sensation of speed and driving skills. Developed for gaming and other entertainment purposes, VR use in other sectors is increasing.

## Application of VR

- **Training:** One of the most obvious is the use of VR in employee training. While this currently requires the use of a headset, it can also be done onsite or at home. The ability to put an employee in other people's shoes (whether those of a co-worker or customer) delivers a unique experience that isn't feasible otherwise. As the technology improves, this will become a valuable tool in all corporate training, including situations that require complex decision-making. VR makes sense in education. Imagine an immersive experience in history or science, for example. As technology progresses and our attention spans decrease, we will continue to expect well-rounded experiences when learning anything new.
- **Travel:** Hotels can take you inside their property, so you know what to expect. VR can be beneficial for high-end travel (e.g., honeymoons or luxury resorts). For the user, they'd see (and feel) the location from their perspective



instead of watching an online video or looking at 2D photos.

- **Real Estate:** Developers can move beyond 3D models to simulate life inside their new development. VR would work both for homes and commercial spaces. Also, co-working spaces can use VR to put the prospective tenant inside the space before you join.
- **Healthcare:** There are many uses for healthcare practitioners, researchers and patients. Imagine using VR to help patients with disorders such as anxiety or anorexia. It would be invaluable in medical school to help students learn how to deal with situations that may arise when they become doctors (empathy training, for example). VR is already in use for surgical training.
- **Retail:** Retailers can help potential consumers put themselves in situations where they can “try on” clothes or objects and get a sense of how they interact with an environment. For example, a bride-to-be could try a wedding dress and place it in an actual wedding environment. VR is different from AR, where you stay in your current reality.
- **Military:** VR is already a valuable tool in simulations for combat, confrontations and the like. It can replace expensive and sometimes dangerous real-life exercises. The ability to change scenarios makes it attractive for all branches of the military and the defense industry.
- **Entertainment:** The ability to provide immersive experiences will transform entertainment. Gaming and Hollywood will increasingly provide users and viewers with the ability to go from passive to active. Consumers will interact with stories in a highly personalized way (should they wish to). The ability to choose your own POV in a

game or movie will continue to provide new forms of engagement.

Other use cases include:

- **Architecture:** VR can render different levels of detail that are important in early-stage design. Architects can create an immersive experience to visualize massing and spatial relationships. Other uses can show how light will affect the proposed space, based on window placement.
- **Art:** VR as a tool for fine art is a staple for artists who aim to push limits. Multimedia artists all over the world are already deeply involved in immersive experiential art forms. Laurie Anderson, a pioneer since the 1970s, was awarded the 74<sup>th</sup> Venice International Film Festival for her work, *The Chalkroom*.
- **Aviation:** Realistic cockpits with VR technology are used to train commercial pilots in training programs incorporating live instruction with virtual flight.
- **Aerospace:** Lockheed Martin builds its F-35 plane with virtual reality technology. In addition to design, engineers now use VR glasses to inspect planes. VR enables engineers to work with up to 96 percent accuracy at a 30 percent faster rate.
- **Conference Rooms:** Users equip themselves with a headset that simulates a meeting room, complete with colleagues and furniture. Motion tracking, combined with digital avatars, allows users to control their digital selves and employ gestures. Even those without headsets use webcam video feeds superimposed on the virtual conference room.
- **Data Visualization:** Engineering and scientific data visualization have profited for years from VR. New display

technology has aroused interest in everything from weather models to molecular visualization.

- **Dining:** Project Nourished replicates eating by manipulating taste, smell, vision, sound, and touch. People experience the virtual as a gourmet meal. The process uses a VR headset, an aroma diffuser, a system that emulates chewing sounds, a rotating utensil and tasteless, 3D-printed food. The project aims to maximize the practical and therapeutic qualities of beverages, medicine and food while limiting natural resource use.
- **Education:** The use of traditional instruction mediums and textbooks is often ineffective for students with special needs. With the introduction of VR, students have become more responsive and engaged. At Charlton Park Academy in London, teachers use immersive technology to address their students' unique needs better.
- **Fashion:** You can find Dior's VR store on its French website. The brand offers shoppers a 3D, 360-degree e-commerce experience. Users virtually browse the store's offerings, zoom in on preferred items and purchase them online.
- **Gaming:** Say "virtual reality," and gaming is the application people think of first. According to the Entertainment Software Association figures reported in March 2020, 73 percent of the 169 million gamers in the U.S. reported owning a gaming console, while 29 percent said they had a VR capable system.
- **Manufacturing:** Designers and engineers easily experiment with the build and look of vehicles before commissioning expensive prototypes with VR. Brands such as Jaguar and BMW use the technology for early design and engineering

reviews. Virtual reality saves the car industry millions by reducing the number of prototypes built per vehicle line.

- **Journalism:** Immersive journalism allows the first-person experience of events or situations described in documentary films and news reports. The Weather Channel uses mixed reality to help communicate everything from wildfires to tornados to flooding.
- **Law Enforcement:** With the advent of VR goggles, virtual reality training has been a boon for law enforcement training. Incident training is realistic and helps prepare officers for everyday situations.
- **Marketing and Advertising:** Virtual reality for marketing allows organizations to bridge the gap between experience and action. VR changes the dynamic between consumers and brands since people seek VR experiences, such as those of Toms Shoes and The North Face.
- **Museums:** Through a mobile phone, projector, headset or web browser, visitors experience locations that would have been unreachable in the recent past. At the National Museum of Natural History in Paris, a permanent VR installation allows visitors to explore different animal species and their links. The exhibit simulates real experiences of interacting or observing animals in their natural habitats.
- **Religion:** There's even an app to experience God. Believe VR and The Virtual Reality Church make it possible for people to worship in depth wherever they are. VR Church became extremely popular during pandemic shutdowns.
- **Social Media:** VR allows people to make connections in a more meaningful way. VRChat gives the power of creation to its community with a wide selection of social VR experiences. Users can hang out, play and chat with

spatialized 3D audio, multiplayer VR games, virtual space stations, and expressive lip-synced avatars.

- **Sports:** VR is a training aid in many sports such as cycling, skiing, golf and gymnastics. At least three college programs—Auburn University, Vanderbilt University and the University of Arkansas—and multiple NFL teams use virtual reality systems.

## What Is Augmented Reality (AR)?

Augmented reality (AR) is an enhanced version of the real physical world that is achieved through the use of digital visual elements, sound, or other sensory stimuli and delivered via technology. It is a growing trend among companies involved in mobile computing and business applications in particular.

Amid the rise of [data collection](#) and analysis, one of augmented reality's primary goals is to highlight specific features of the physical world, increase understanding of those features, and derive smart and accessible insight that can be applied to real-world applications.

Augmented reality (AR) involves overlaying visual, auditory, or other sensory information onto the real world to enhance one's experience.

Retailers and other companies can use augmented reality to promote products or services, launch novel marketing campaigns, and collect unique user data.

Unlike virtual reality, which creates its own cyber environment, augmented reality adds to the existing world as it is.

Augmented reality continues to develop and become more pervasive among a wide range of applications. Since its conception, marketers and [technology](#) firms have had to battle the perception that augmented reality is little more than a [marketing](#) tool. However, there is evidence that consumers are beginning to derive tangible benefits from this technology and expect it as part of their purchasing process.

Some experts have long speculated that [wearable](#) devices could be a breakthrough for augmented reality. [Smartphones](#) and tablets show a tiny portion of the user's landscape, but smart eyewear, for example, may

provide a more complete link between real and virtual realms if it develops enough to become mainstream.

### Example

Some [early adopters](#) in the retail sector have developed AR technologies designed to enhance the consumer shopping experience. Augmented reality has been incorporated into store catalog apps, which allows consumers to visualize what different products would look like in different environments. For example, when buying furniture, shoppers point the camera to the appropriate room and the product will appear in the foreground.

Augmented reality's benefits can also extend to the [healthcare sector](#), where it can play a much bigger role. AR apps enable users to see highly detailed, 3D images of different body systems when they hover their mobile device over a target image. This use of AR has become a powerful learning tool for training medical professionals.

## Augmented Reality vs. Virtual Reality

Augmented reality and virtual reality are often confused, so let's clarify. Augmented reality uses the existing real-world environment and puts virtual information—or even a virtual world—on top of it to enhance the experience. For example, think of Pokémon Go, where users are searching in their real-life neighborhoods for animated characters that pop up on their phone or tablet. In the NFL, broadcasters use AR technology to better analyze plays.

In contrast, virtual reality immerses users into an entirely different environment, typically a virtual one created and rendered by computers. For example, virtual reality users may be immersed in an animated scene or a digital environment. Virtual reality can also be used to photograph an actual local location and embed it in a VR app. Through a virtual reality headset, someone can walk around Italy as if they were actually there.

| AR                                                           | VR                                                   |
|--------------------------------------------------------------|------------------------------------------------------|
| The system augments the real-world scene                     | Completely immersive virtual environment             |
| In AR User always have a sense of presence in the real world | In VR, visual senses are under control of the system |
| AR is 25% virtual and 75% real                               | VR is 75% virtual and 25% real                       |

|                                                                                                                  |                                                                                                                       |
|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| This technology partially immerses the user into the action                                                      | This technology fully immerses the user into the action                                                               |
| AR requires upwards of 100 Mbps bandwidth                                                                        | VR requires at least a 50 Mbps connection                                                                             |
| No AR headset is needed.                                                                                         | Some VR headset device is needed.                                                                                     |
| With AR, end-users are still in touch with the real world while interacting with virtual objects nearer to them. | By using VR technology, VR user is isolated from the real world and immerses himself in a completely fictional world. |
| It is used to enhance both real and virtual worlds.                                                              | It is used to enhance fictional reality for the gaming world.                                                         |

## What Is Artificial Intelligence (AI)?

Artificial intelligence (AI) refers to the simulation of human intelligence in machines that are programmed to think like humans and mimic their actions. The term may also be applied to any machine that exhibits traits associated with a human mind such as learning and problem-solving.

The ideal characteristic of artificial intelligence is its ability to rationalize and take actions that have the best chance of achieving a specific goal. A subset of artificial intelligence is [machine learning](#) (ML), which refers to the concept that computer programs can automatically learn from and adapt to new data without being assisted by humans. [Deep learning](#) techniques enable this automatic learning through the absorption of huge amounts of unstructured data such as text, images, or video.

- Artificial intelligence (AI) refers to the simulation or approximation of human intelligence in machines.
- The goals of artificial intelligence include computer-enhanced learning, reasoning, and perception.
- AI is being used today across different industries from finance to healthcare.
- Weak AI tends to be simple and single-task oriented, while strong AI carries on tasks that are more complex and human-like.
- Some critics fear that the extensive use of advanced AI can have a negative effect on society.

## Applications of Artificial Intelligence

The applications for artificial intelligence are endless. The technology can be applied to many different sectors and industries. AI is being tested and used in the healthcare industry for dosing drugs and doling out different treatments tailored to specific patients, and for aiding in surgical procedures in the operating room.

Other examples of machines with artificial intelligence include computers that play chess and [self-driving cars](#). Each of these machines must weigh the consequences of any action they take, as each action will impact the end result. In chess, the end result is winning the game. For self-driving cars, the computer system must account for all external data and compute it to act in a way that prevents a collision.

Artificial intelligence also has [applications in the financial industry](#), where it is used to detect and flag activity in banking and finance such as unusual debit card usage and large account deposits—all of which help a bank's fraud department. Applications for AI are also being used to help streamline and make trading easier. This is done by making supply, demand, and pricing of securities easier to estimate.

## Types of Artificial Intelligence

Artificial intelligence can be divided into two different categories: weak and strong. [Weak artificial intelligence](#) embodies a system designed to carry out one particular job. Weak AI systems include video games such as the chess example from above and personal assistants such as Amazon's Alexa and Apple's Siri. You ask the assistant a question, and it answers it for you.

[Strong artificial intelligence](#) systems are systems that carry on the tasks considered to be human-like. These tend to be more complex and complicated systems. They are programmed to handle situations in which they may be required to problem solve without having a person intervene. These kinds of systems can be found in applications like self-driving cars or in hospital operating rooms.

## What Are the 4 Types of AI?

Artificial intelligence can be categorized into one of four types.

- **Reactive AI** uses algorithms to optimize outputs based on a set of inputs. Chess-playing AIs, for example, are reactive systems that



optimize the best strategy to win the game. Reactive AI tends to be fairly static, unable to learn or adapt to novel situations. Thus, it will produce the same output given identical inputs.

- **Limited memory AI** can adapt to past experience or update itself based on new observations or data. Often, the amount of updating is limited (hence the name), and the length of memory is relatively short. Autonomous vehicles, for example, can "read the road" and adapt to novel situations, even "learning" from past experience.
- **Theory-of-mind AI** are fully-adaptive and have an extensive ability to learn and retain past experiences. These types of AI include advanced chat-bots that could pass the [Turing Test](#), fooling a person into believing the AI was a human being. While advanced and impressive, these AI are not self-aware.
- **Self-aware AI**, as the name suggests, become sentient and aware of their own existence. Still in the realm of science fiction, some experts believe that an AI will never become conscious or "alive".

### **The basic components of AI include;**

- Learning
- Reasoning
- Problem-solving
- Perception
- Language understanding

### **How Is AI Used Today?**

AI is used extensively across a range of applications today, with varying levels of sophistication. Recommendation algorithms that suggest what you might like next are popular AI implementations, as are chatbots that appear on websites or in the form of smart speakers (e.g., Alexa or Siri). AI is used to make predictions in terms of weather and financial forecasting, to streamline production processes, and to cut down on various forms of redundant cognitive labor (e.g., tax accounting or editing). AI is also used to play games, operate autonomous vehicles, process language, and much, much, more.

### **What is Ambient Intelligence?**

Ambient Intelligence (Aml) is a technology that combines Artificial Intelligence (AI), the Internet of Things (IoT), Big Data, Pervasive-Ubiquitous Computing, Networks, and Human Computer Interaction (HCI).

The technology provides the ideal marriage of IoT and AI. [IoT devices](#) and sensors implanted in the user's surroundings (such as their home and office) will collect contextual data and utilize AI to forecast the user's requirements.

Ambient intelligence (Aml) is a new technology that intends to introduce pervasive computing, artificial intelligence (AI) sensors, and sensor networks into our daily life. This technology is human-centric because it is extremely sensitive to the presence of humans in its surroundings.

It is a sophisticated artificial intelligence system that recognizes and responds to human presence. Siri or Alexa are actual examples of [ambient intelligence](#) since they reply to you once they detect your voice.

Eli Zelkha and his colleagues pioneered the notion of ambient intelligence in the late 1990s. They first wanted to design technologies that would collaborate with people to make their lives simpler. The gadgets would collect and store data and intelligence in linked devices, such as those powered by IoT.

Ambient intelligence interfaces provide intriguing difficulties in terms of how to communicate Aml information to the user and how the user should interact with Aml applications. While many Aml apps employ implicit human-computer interaction and hence do not require explicit user input, the majority do demand some active user engagement.

As interesting as Aml systems are, they have also raised privacy issues. Many individuals, understandably, will be uncomfortable with intelligent spaces gathering data without their permission.

AI still suffers from prejudice, and user information falling into the wrong hands might have catastrophic consequences. Navigating the ethical problems of Ambient Intelligence will be difficult.

## **Factors of Aml**

## Usability

First and foremost, contact with Ambient Intelligence should be unobtrusive, with no steep learning curve. Many firms have embraced user-friendly design ideas in their products, influenced by Apple's design philosophy.

Aml will take it a step farther. Aml settings should not necessitate the use of a manual. When you enter into an Aml environment, it should just function.

### 2. Technical Suitability

Furthermore, Aml must function reliably within the restrictions of cutting-edge technology. Accuracy, capacity, and fail-safe mechanisms for all hardware and software components must be taken into account.

In early 2016, [Microsoft Tay](#) showed how artificial intelligence (AI) may abruptly go wild. We still have a long way to go before completely comprehending how sophisticated AI systems operate.

### 3. Belief and Confidence

Furthermore, Aml must acquire the confidence and trust of users before it can be broadly implemented. Former Baidu chief scientist Andrew Ng stated that AI is the new electricity. The IoT infrastructure that provides Aml services will be similar to the energy infrastructure.

To acquire the trust of the general public, Aml must prioritize [risk management](#), [data security](#), and system dependability. As Ambient Intelligence gets more common and powerful, privacy is becoming increasingly important.

The capacity to uniquely identify and monitor people is fundamental in Aml. To acquire consumers' confidence, we must explicitly establish privacy regulations in both public and private spaces.

#### 4. Social and Economic Consequences

Aml should be capable of improving the social relationships of all individuals in the environment. One example is the employment of Aml to improve healthcare access, geriatric care, and social support for the elderly.

Last but not least, the economic impact on stakeholders could not be overstated. On the one hand, it will undoubtedly boost societal production. On the other hand, it will permanently alter the work duties of many employees in the affected industries.

### **Applications of Ambient Intelligence**

In the next 10 years, all of the aforementioned technologies will be integrated in our homes, grocery shops, businesses, hospitals, and transportation systems, ushering in an Ambient Intelligence (Aml) revolution. The following are some of the most major [applications of Ambient Intelligence](#) are :-

## 1. Education

The growth of technology has resulted in significant changes in the Teacher-Student interaction, which no longer takes place face-to-face and may rely on [communication technologies](#) that are always inadequate substitutes for our rich communication processes.

Aml appears to be one feasible approach to coping with some of the neglected difficulties that emerge from this rapid (and often uncontrollable) technological progress. One of these issues is the existing internet communication technologies' lack of contextual information, which is often perceived as cold and impersonal.

Teachers and students who used to talk face-to-face now communicate through forums and text-based discussions, which makes communication significantly worse.

## 2. Robotics

Robots have long been an element of the Artificial Intelligence fantasy. Traditionally, in the near future, each individual would supposedly have a robotic companion, most likely in humanoid form, acting as a multi-

faceted personal helper that would attend to all of our requirements. Research and technological advancement appear to be moving in a different path.

The most plausible future is one in which an ecology of agents, often our traditional appliances and hardware, communicate and coordinate to serve us via some form of enabling middleware. Robots that behave similarly to us, sharing our environment and moving fluidly through it while offering services, can be imagined.

### 3. Health

The healthcare industry is presently one of the most technologically advanced. As a result, it is not surprising that many different projects may be found in this category. There are two major developments to note: the provision of healthcare services at home and the support of healthcare in the hospital.

The primary goal in all instances is to reduce healthcare expenses while making it more accessible, tailored, and available. Most initiatives in the home scenario are aimed towards assisting the elderly, or providing them with greater freedom and quality of life.

In the healthcare institution, one can plan projects to deal with extremely specific sorts of scenarios (for example, Alzheimer patients) as well as more general ones, such as the employment of discrete hidden Markov models for categorizing the activities of healthcare practitioners.

#### 4. Transportation

Given that we spend a significant portion of our life traveling back and forth, transportation is another field in which the adoption of Ambient Intelligence might result in several benefits.

Buses, and vehicles may be outfitted with technology that can offer essential knowledge about how the system is working and identification of possible changes by more effectively using the system, which can result in an improvement in people's experiences.

GPS-based spatial positioning of public transportation vehicles and services; vehicle identification and image processing to improve transportation efficiency and safety, or to discover circumstances of interest in congested areas , among many more.

**Also Read |** [Conversational Intelligence: Concept, Importance and Tools](#)

#### 5. Services for Emergencies

Emergency services and their organization form a complicated and important environment in which Ambient Intelligence applications might be found. The key goals here are to provide mission-critical knowledge in real time and to coordinate the many players.

Another goal is to discuss the prospects of Aml in emergency services, including their possible role in the quick construction of virtual response teams, as well as the assistance for communication and coordination of such teams' members, even when they span different nations.

## 6. Workplace

Ambient Intelligence has the potential to significantly minimize the occurrence of accidents in workplaces such as construction sites. Many mishaps are caused by a failure to follow safety requirements.

Human monitors cannot be present throughout the day. Aml, on the other hand, has a constant bird's-eye perspective of the surroundings and the personnel. It can provide rapid notifications to the management or worker if a safety regulation is broken.

## 7. Online to Offline Commerce

For several years, retailers have used wireless technology such as iBeacon to reach out to smartphone customers while they are shopping in physical locations.

On the other side, as part of their Online to Offline (O2O) initiatives, E-Commerce behemoths like Amazon and Alibaba begin to investigate the notion of unmanned supermarkets (Amazon Go and Tao Cafe, for example).



These stores make extensive use of Aml technology. They intend to provide a personalized 24/7 shopping experience in physical locations that is comparable to internet purchasing.

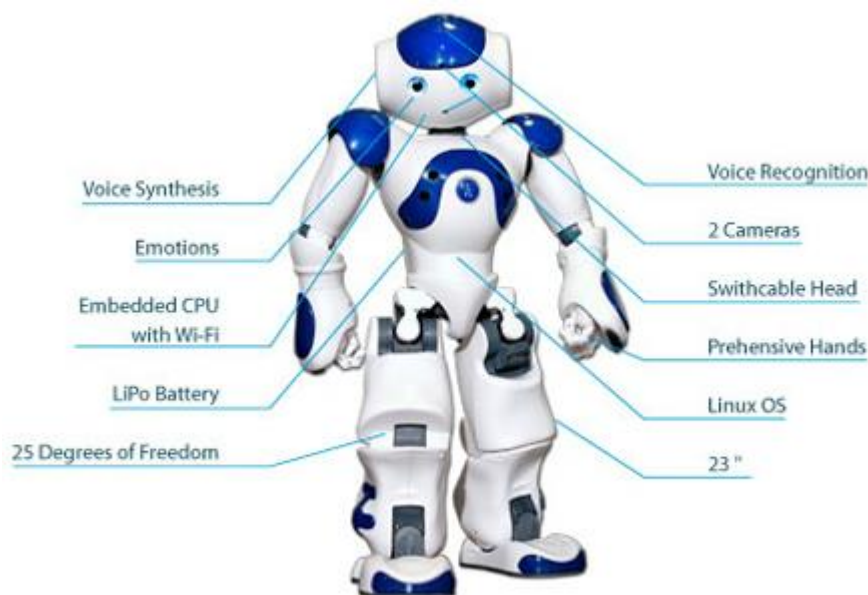
The ambient future is taking shape, and many of its early-generation goods and services provide considerable advantages. But, as with AI and other technologies that are infiltrating our personal and professional lives, one crucial message for humans is clear: we will all need to grow better at, and more comfortable with, weighing advantages and dangers.

## Robots

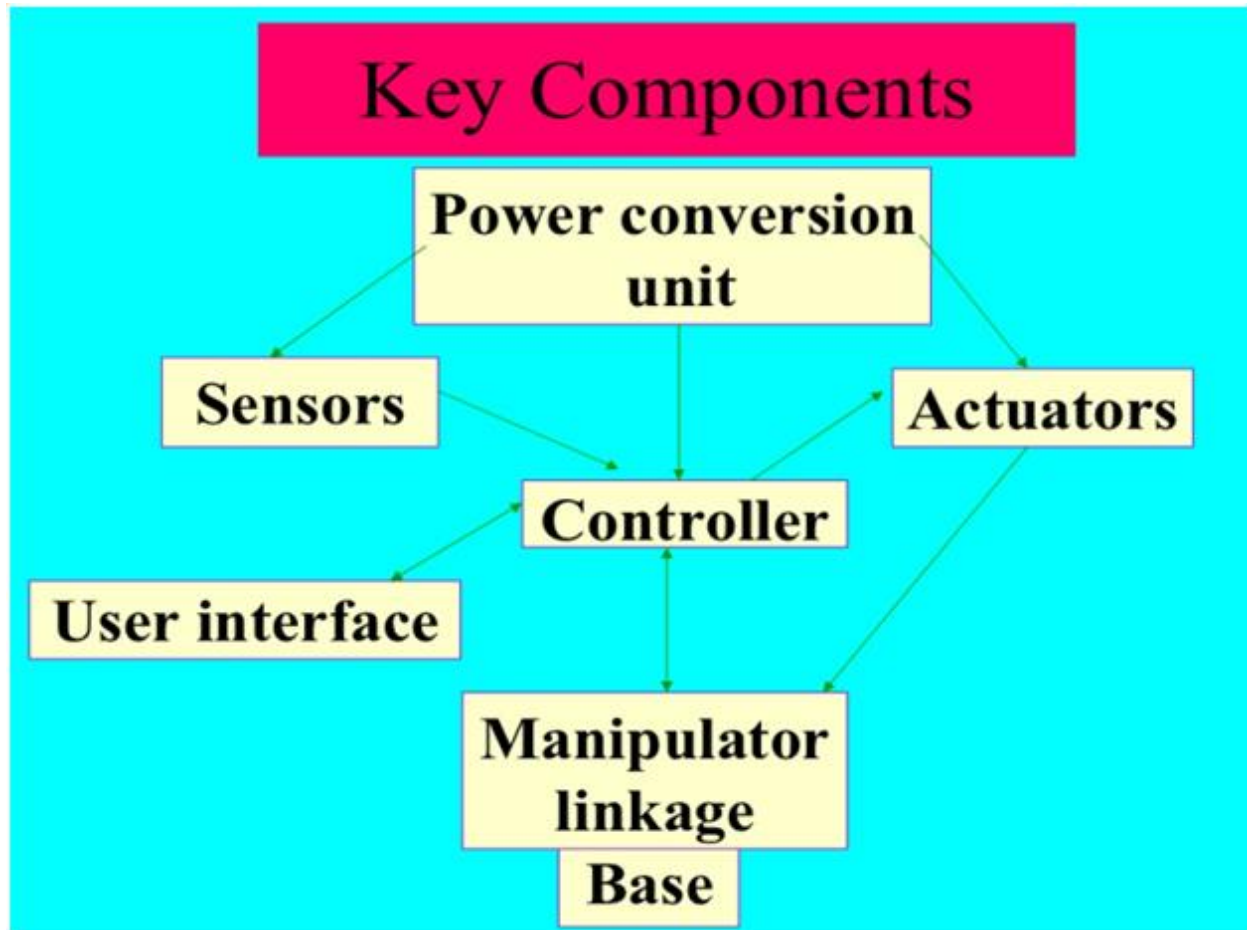
A robot is a **type of automated machine that can execute specific tasks with little or no human intervention and with speed and precision**. The field of robotics deals with robot design, engineering and operation.

## Components of Robot

Consider the robot structure showing different components of robots are:



Consider the key components of robotics are:-



- **Power Supply** - The working power to the robot is provided by batteries, hydraulic, solar power, or pneumatic power sources.
- **Actuators** - Actuators are the energy conversion device used inside a robot. The major function of actuators is to convert energy into movement.
- **Electric motors (DC/AC)**- Motors are electromechanical component used for converting electrical energy into its equivalent mechanical energy. In robots motors are used for providing rotational movement.
- **Sensors** - Sensors provide real time information on the task environment. Robots are equipped with tactile sensor it imitates the mechanical properties of touch receptors of human fingerprints and a vision sensor is used for computing the depth in the environment.

- **Controller** - Controller is a part of robot that coordinates all motion of the mechanical system. It also receives an input from immediate environment through various sensors. The heart of robot's controller is a microprocessor linked with the input/output and monitoring device. The command issued by the controller activates the motion control mechanism, consisting of various controller, actuators and amplifier.

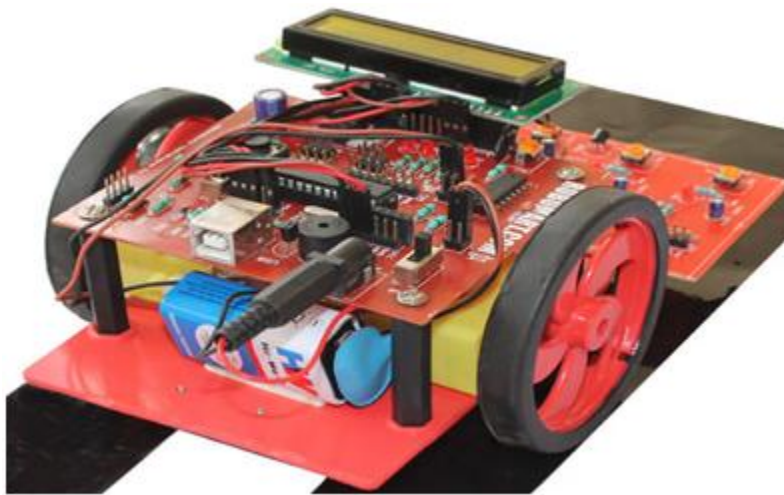
## Types of Robots

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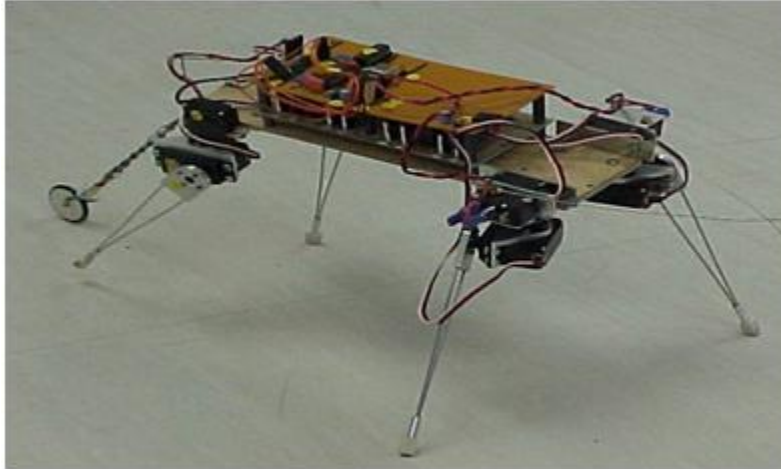
### 1) Mobile Robots

Mobile robots are able to move from one location to another location using locomotion. It is an automatic machine that is capable of navigating an uncontrolled environment without any requirement of physical and electromechanical guidance devices. Mobile Robots are of two types:

**(a) Rolling robots** - Rolling robots require wheels to move around. They can easily and quickly search. But they are only useful in flat areas.



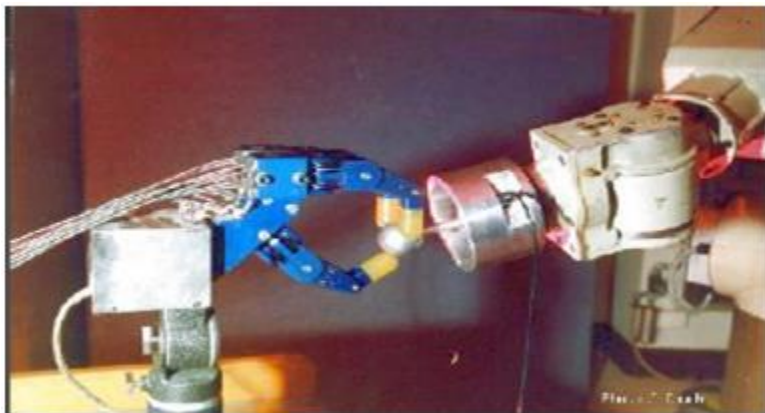
**(b) Walking robots** - Robots with legs are usually used in condition where the terrain is rocky. Most walking robots have at least 4 legs.



## 2) Industrial Robots

Industrial robots perform same tasks repeatedly without ever moving. These robots are working in industries in which there is requirement of performing dull and repeated tasks suitable for robot.

An industrial robot never tires, it will perform their work day and night without ever complaining.



## 3) Autonomous Robots

Autonomous robots are self-supported. They use a program that provides them the opportunity to decide the action to perform depending on their surroundings.

Using artificial intelligence these robots often learn new behavior. They start with a short routine and adapt this routine to be more successful in a task they perform. Hence, the most successful routine will be repeated.



## 4) Remote Controlled Robots

Remote controlled robot used for performing complicated and undetermined tasks that autonomous robot cannot perform due to uncertainty of operation.

Complicated tasks are best performed by human beings with real brainpower. Therefore a person can guide a robot by using remote. Using remote controlled operation human can perform dangerous tasks without being at the spot where the tasks are performed.

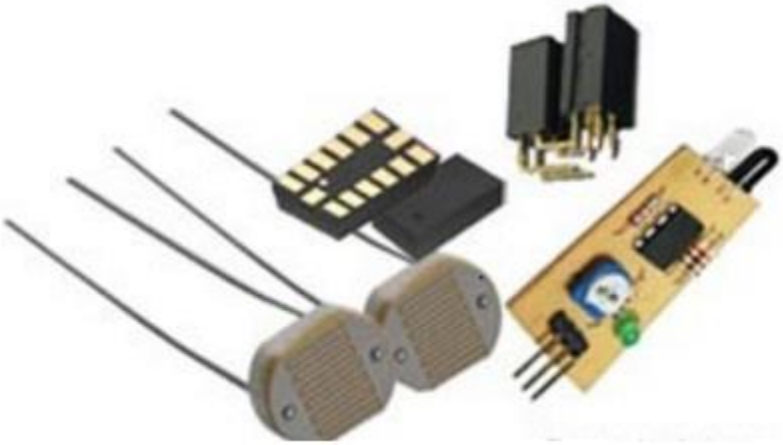
Let's see a NASA robot designed to explore volcanoes via remote control:





# Types of Robot Sensors

There are different type of sensors are available to choose from and the characteristics of sensors are used for determining the type of sensor to be used for particular application.



## 1) Light Sensor

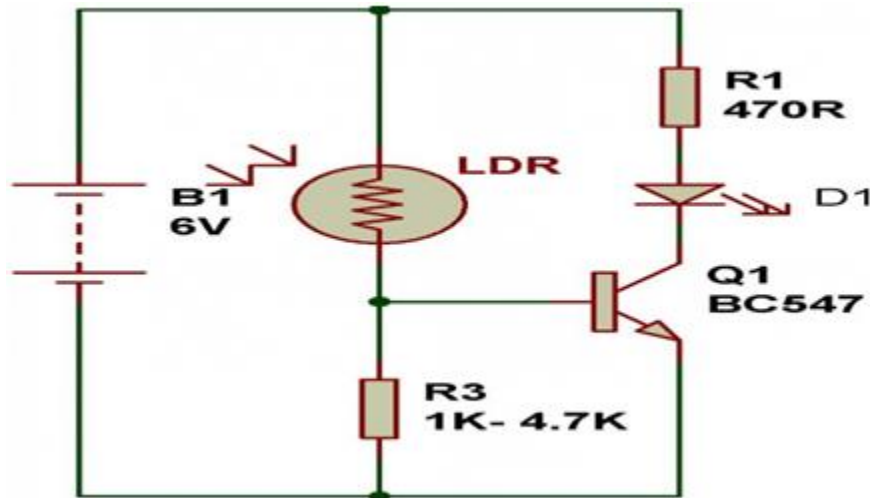
Light sensor is a transducer used for detecting light and creates a voltage difference equivalent to the light intensity fall on a light sensor.

The two main light sensors used in robots are **Photovoltaic cells** and **Photo resistor**. Other kind of light sensors like phototransistors, phototubes are rarely used.

The type of light sensors used in robotics are:

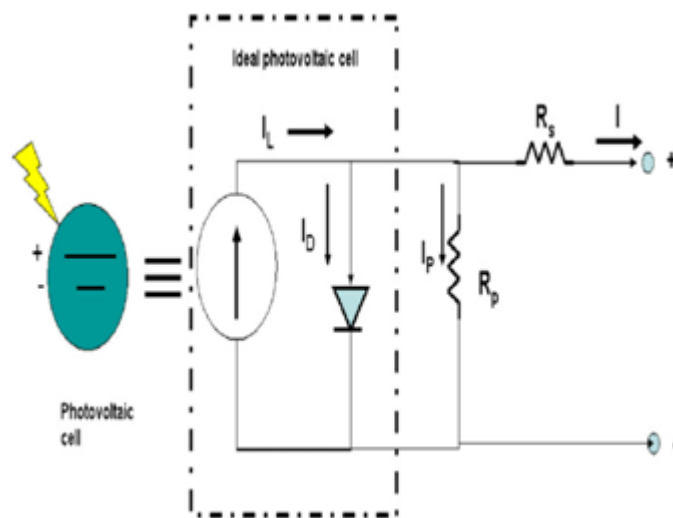
**Photo resistor** - It is a type of resistor used for detecting the light. In photo resistor resistance varies with change in light intensity. The light falls on photo resistor is inversely proportional to the resistance of the photo resistor. In general photo resistor is also called as Light Dependent Resistor (LDR).

Consider the circuit diagram of Photo resistor sensor:



**Photovoltaic Cells** - Photovoltaic cells are energy conversion device used to convert solar radiation into electrical electric energy. It is used if we are planning to build a solar robot. Individually photovoltaic cells are considered as an energy source, an implementation combined with capacitors and transistors can convert this into a sensor.

Consider the circuit diagram of photovoltaic cell is,



## 2) Proximity Sensor

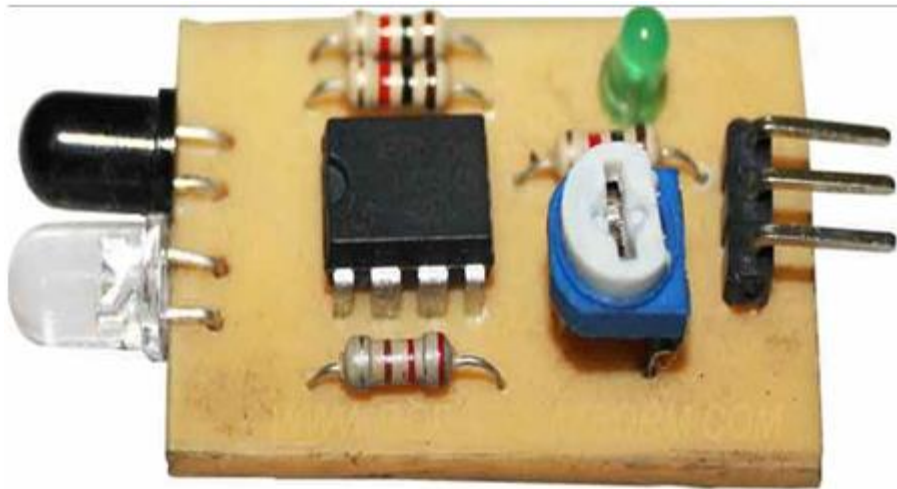
Proximity sensor can detect the presence of nearby object without any physical contact. The working of a proximity sensor is simple. In proximity sensor transmitter transmits an electromagnetic radiation and receiver receives and analyzes the return signal for

interruptions. Therefore the amount of light receiver receives by surrounding can be used for detecting the presence of nearby object.

Consider the types of proximity sensors used in robotics are:-

**Infrared (IR) Transceivers** - In IR sensor LED transmit the beam of IR light and if it find an obstacle then the light is reflected back which is captured by an IR receiver.

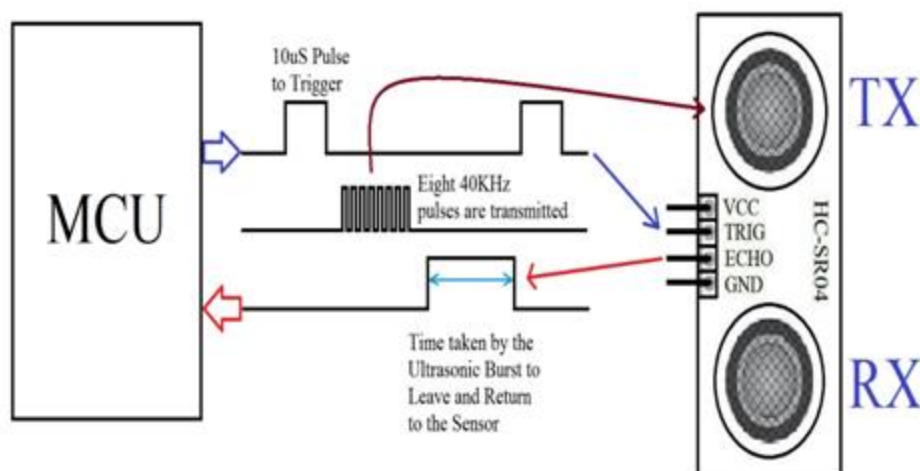
Consider the PCB board layout of IR Transceiver circuit:



**Ultrasonic Sensor** - In ultrasonic sensors high frequency sound waves is generated by transmitter, the received echo pulse suggests an object interruption.

In general ultrasonic sensors are used for distance measurement in robotic system.

Consider the interfacing of ultrasonic sensor with Microcontroller unit:





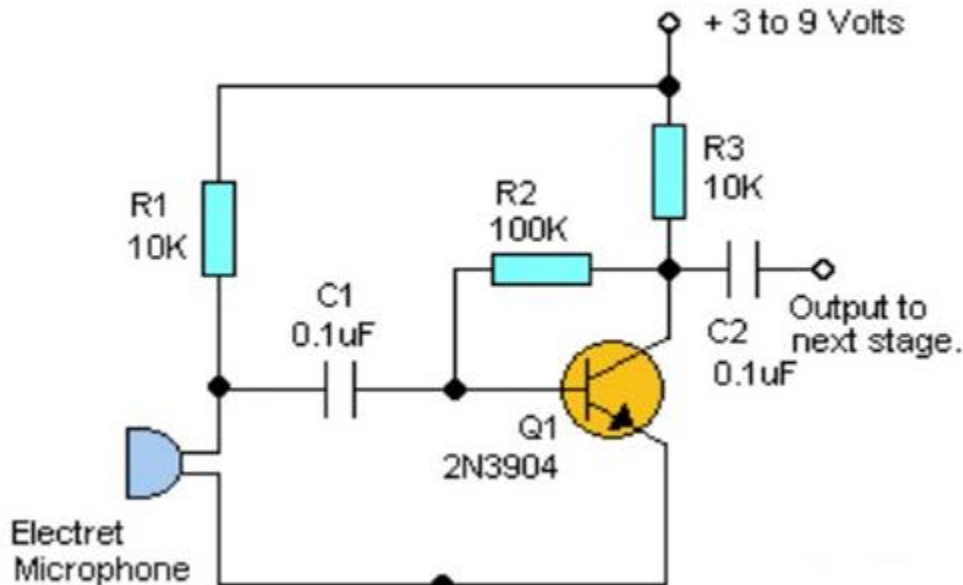
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### 3) Sound Sensor

Sound sensors are generally a microphone used to detect sound and return a voltage equivalent to the sound level. Using sound sensor a simple robot can be designed to navigate based on the sound receives.

Implementation of sound sensors is not easy as light sensors because it generates a very small voltage difference which will be amplified to generate measurable voltage change.

Consider the sound sensor based switching circuit:



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### 4) Temperature Sensor

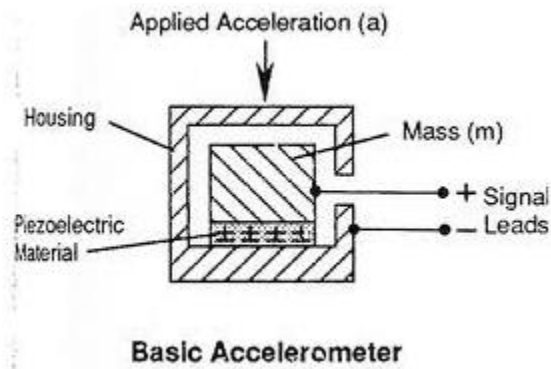
Temperature sensors are used for sensing the change in temperature of the surrounding. It is based on the principle of change in voltage difference for a change in temperature this change in voltage will provide the equivalent temperature value of the surrounding.

Few generally used temperature sensors IC?s are TMP35, TMP37, LM34, LM35, etc.

Consider the temperature sensor pin diagram description is,



Consider the schematic diagram of basic accelerometer:



## Applications of Robots

### 1. Security

Imagine if all the security guards are robots? Even thieves would be scared! That's why robots are being proposed as security agents as they can protect humans, and they wouldn't be in danger like human security guards would be. Currently, robotics companies are working on pairing robot guards with human security consultants. A very famous company in this field is **Knightscope** in the United States that has autonomous security robots capable of assisting human security guards with real-time, actionable intelligence. These robots can help with crimes such as armed robberies, burglaries, domestic violence, fraud, hit, and runs, etc.

### 2. Space Exploration

There are many things in space that are very dangerous for astronauts to do. Humans can't roam on Mars all day to collect soil samples or work on repairing a spaceship from the outside while it's in deep space! In these situations, robots are a great choice because there are no chances for the loss of human life then. So space institutions like NASA frequently use robots and autonomous vehicles to do things that humans can't. For example, Mars Rover is an autonomous robot that travels on Mars and takes pictures of Martian rock formations that are interesting or important and then sends them back on Earth for the NASA scientists to study.

### 3. Entertainment

Robots are also a big draw in the entertainment industry. While they cannot exactly become actors and actresses, they can be used behind the sets in movies and serials to manage the camera, provide special effects, etc. They can be used for boring repetitive tasks that are not suitable for a human as cinema is, after all, a creative industry. Robots can also be used to do stunt work that is very dangerous for humans but looks pretty cool in an action movie. Theme parks like Disney World are also using autonomous robots to enhance the magical experience of their customers.

### 4. Agriculture

Agriculture is the sector that is the basis of human civilization. However, agriculture is also a seasonal sector that is dependent on ideal weather conditions optimal soil, etc. Moreover, there are many repetitive tasks in agriculture that are just a waste of farmer's time and can be performed more suitably by robots. These include seeding, weed control, harvesting, etc. Robots are usually used for harvesting the crops which allow farmers to be more efficient. An example of a robot that is used to remove weeds in farms is the Ecorobotix. It is powered by solar energy and can be used to target and spray weeds using a complex camera system.

### 5. Health Care

Robots have changed healthcare a lot. And all for the better! They can help doctors in performing operations more precisely, be used as prosthetic limbs, provide therapy to patients, etc. The possibilities are limitless. One example of this is the **da Vinci robot** that can help surgeons in performing complex surgeries relating to the heart, head, neck, and other sensitive areas. There are other robotic devices that are created like exoskeletons that can be used to provide additional support for people undergoing rehabilitation after spinal injuries, strokes, etc.

### 6. Underwater Exploration

Robots are a great option for exploring places that humans cannot reach easily, like the depths of the ocean! There is a lot of water pressure deep in the ocean which means humans cannot go that down and machines such as submarines can only go to a certain depth as well. A deep underwater is a mysterious place that can finally be explored using specially designed robots. These robots are

remote-controlled, and they can go into depths of the ocean to collect data and images about the aquatic plant and animal life.

## 7. Food Preparation

Don't want to cook? Don't worry, there are robots that even can cook and create complete meals for you! These robot chefs can create food using hundreds of different recipes. All humans need to do is choose the recipe they want and provide the robot with pre-packaged containers of all the ingredients that are needed for that recipe. The robot can then cook the food on its own. Moley Robotics is one such robotics company that has created a robotic kitchen with a robot that can cook like a master chef! So no worries if you can't cook food. Because now a robot can!

## 8. Manufacturing

There are many repetitive and common tasks in the manufacturing industry that don't require any usage of the mind like welding, assembly, packing, etc. These tasks can be easily done by robots while leaving the mentally challenging and creative tasks to humans. These robots can be trained to perform these repetitive and monotonous tasks with precision under the guidance and supervision of a human. This option is also best for the manufacturing processes that are dangerous and may be harmful to humans.

## 9. Military

Robots also have many applications in the military. They can be used as drones to keep surveillance on the enemy, they can also be used as armed systems to attack the opposing forces or as Medicare agents to help friendly forces. Some of the popular robots used in the Military sector include MAARS (Modular Advanced Armed Robotic System) which looks like a tank and contains tear gas and lasers to confuse enemies and even grenade launcher for desperate situations. DOGO is also a tactical combat robot that has a camera for spying on the activities of the enemy and a 9-millimeter pistol for emergency situations!

## 10. Customer Service

There are robots that are developed to look exactly like humans for cosmetic purposes. These robots are primarily used in the field of customer service in high visibility areas to promote robotics. One such example is Nadine, a humanoid robot in Singapore that can recognize people from previous visits, make eye

contact, shake hands, continue chatting based on previous meetings, etc. Another such customer service robot is Junko Chihira in Japan, a humanoid robot working at the tourist information center in Aqua City Odaiba, a shopping center on Tokyo's waterfront.

## What Is Bitcoin?

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Bitcoin is a cryptocurrency which is based on block chain technology.

Blockchain is the technology that enables the existence of cryptocurrency (among other things). Bitcoin is the name of the best-known cryptocurrency, the one for which blockchain technology, as we currently know it, was created. A cryptocurrency is a medium of exchange, such as the US dollar, but is digital and uses cryptographic techniques and its protocol to verify the transfer of funds and control the creation of monetary units.

What is blockchain technology?

A blockchain is a decentralized ledger of all transactions across a peer-to-peer network. Using this technology, participants can confirm transactions without a need for a central clearing authority. Potential applications can include fund transfers, settling trades, voting and many other issues.

## What is cryptocurrency?

### Cryptocurrency

**Cryptocurrency** is a medium of exchange, created and stored electronically on the blockchain, using cryptographic techniques to verify the transfer of funds and an algorithm to control the creation of monetary units. Bitcoin is the best known example.



**Has no intrinsic value** in that it is not redeemable for another commodity, such as gold.



**Has no physical form** and exists only in the network.



**Its supply is determined by the protocol**, not a central bank and the network is completely decentralized.

# Blockchain's benefits and unknowns

## Benefits



Increased transparency



Accurate tracking

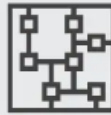


Permanent ledger



Cost reduction

## Unknowns



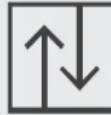
Complex technology



Regulatory implication

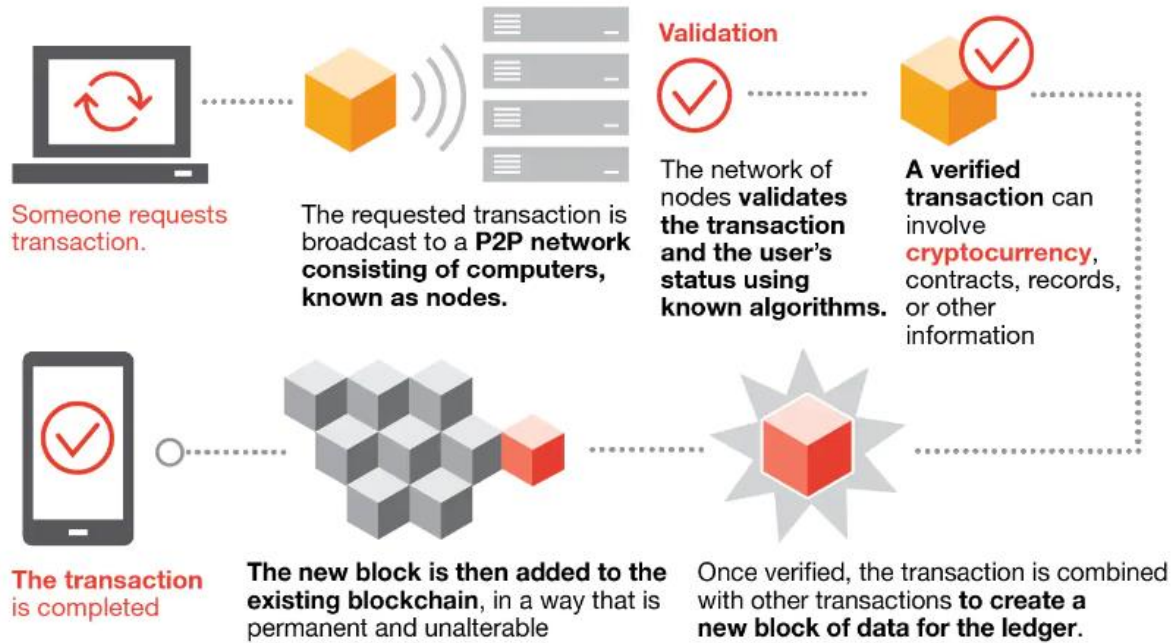


Implementation challenges



Competing platforms

# How blockchain works





# Blockchain's potential applications

## Potential applications



### Automotive

**Consumers could use the blockchain** to manage fractional ownership in autonomous cars.



### Financial services

**Faster, cheaper settlements** could shave billions of dollars from transaction costs while improving transparency.



### Voting

Using a blockchain code, constituents could cast votes via smartphone, tablet or computer, **resulting in immediately verifiable results.**



### Healthcare

**Patients' encrypted health information** could be shared with multiple providers without the risk of privacy breaches.